MILITARY SPECIFICATION

DISPLAY UNIT, FOR ROCKET MANAGEMENT SUBSYSTEM M138

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions, and preparation for delivery for Display Unit, for Rocket Management Subsystem, M138.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-5400	-	Electronic Equipment, Airborne, General Specification for
MIL-C-6781	-	Control Panel: Aircraft Equipment Rack or Console Mounted
MIL-P-7788	-	Panels, Information; Integrally Illuminated Plate, Tags and Bands for Identification of Equipment
MIL-W-63150	-	Weapons and Support Material, Standard Quality Assurance Provisions for
MS25212	-	Control Panel, Console Type, Aircraft Equipment, Basic Dimensions

FSC 1310

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-280	-	Definitions of Item Levels, Item Exchangeability, Models and Related Terms
MIL-STD-704A	-	Aircraft Electrical Power Characteristics
MIL-STD-810C	-	Environmental Methods (with Notice 2 (11))
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein:

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT

9324104	-	DV Interface Control Drawing
9324107	-	Display Unit
9324400	-	DU/OU Interface Specification
A12011785	-	Burn-In Test Procedure
12011837	-	Signal Interface Control Document

PACKAGING

P9324107	-	Operations Unit

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3951	-	Standard Practice for Commercial Practice

(Copies of specifications, standards, handbooks, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Item definition. The Display Unit (DU) is used with four Operations Units to form the M138 Rocket Management Subsystem. It provides the capability while in flight to select, provide data for setting resistance-capacitance (RC) fuzes, and control the firing of 2.75 inch aerial rockets from two or four wing stations. The DU has the capability of managing stores of up to five different rocket types. Rockets are selectable for single, pair, or quadruple firing. Rocket quantities and firing rates are selectable. The DU provides a visual presentation of the number of rockets of each type remaining. Major components of the DU are: Control Panel Assembly: Control Panel Interface Assembly; Central Processor Assembly; Memory Assembly; Built-In-Test Assembly; Input/Output Assembly; Power Supply Assembly and Motherboard Assembly.

3.2 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract and 4.4.

3.3 Materials. Materials shall be in accordance with applicable drawings, specifications and standards.

3.4 Design and construction. The M138 Display Unit shall comply with all requirements specified on Drawing (Dwg) 9324107, all associated drawings, and with all requirements specified in applicable specifications and standards. The DU shall be constructed to meet the requirements of MIL-E-5400, for Class 1A equipment. Because of compact design of the DU Power Supply, a maximum of five wires may be terminated at a terminal.

3.4.1 Interchangeability. All Display Units manufactured to this specification shall be functionally interchangeable as defined in MIL-STD-280. Interchangeability shall exist between units and all replaceable assemblies, subassemblies and parts having the same part number.

3.4.2 Preconditioning (burn-in). The "burn-in" shall be conducted in accordance with, Burn-in Test Procedure, 12011785.

3.4.3 Characteristics.

3.4.3.1 Performance. The performance characteristics of the DU shall be as defined in the following paragraphs.

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3.4.3.1.1 Front panel. All DU controls shall be mounted on the front panel. Controls shall include warhead/fuze type selector switches, range and penetration selector switches, selector switches for controlling rate of fire, firing mode and quantity, five zone arm pushbutton switches, and a push to test switch. A rounds remaining display shall be used to indicate the number of rockets remaining to be fired for each zone.

3.4.3.1.1.1 Warhead/fuze selectors. The DU shall divide rocket tubes in four different launchers into five firing zones. The zones shall be grouped as shown in Figure 1. Each zone must be loaded with rockets having the same warhead/fuze combination with the warhead/fuze selector switch being set accordingly. Each warhead/fuze selector shall have 12 positions with the following legends and associated information.

Switch Position	Legend	Warhead Type	RC Fuze Type	Auto Rate
1	CHF	Chaff	XM439	Fast
2	MPS	Multipurpose RC	XM439	Fast
3	SMK	Smoke RC	XM439	Fast
4	ILM	Illuminating RC	XM439	Fast
5	FLC	Flechette RC	XM439	Fast
6	H17	H17 lb High Explosive	None	Slow
7	WP	White Phosphorous Point Detonating	None	Fast
8	HDP	High Explosive Dual Purpose	None	Fast
9	HRC	High Explosive RC	433	Fast
10	HPX	High Explosive Proximity	None	Slow
11	HPD	High Explosive Point Detonating	None	Fast
12	--	Special Purpose	None	Fast

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3.4.3.1.1.2 Range selectors. The DU shall provide two thumbwheel switches for the selection of range for use with XM439 fuzes. Selectable ranges shall include 0.5 to 6.0 kilometers. Switch settings below 0.5 km shall be interpreted as 0.5 km and settings between 6.1 and 6.9 km shall be interpreted as 6.0 km. An eighth position of the most significant switch shall be included for the automatic range selection. Table 1 shows the time of function for each setting. In the "A" position of the most significant switch, the RMS will accept range data from the Fire Control Computer (FCC). In the absence of or failure of an FCC, the RMS will revert to a default range setting of 3.0 km.

TABLE I. Time of flight for range settings (cont'd).

Range (KM)	Time (See)
0.0	.950
0.1	.950
0.2	.950
0.3	.950
0.4	.950
0.5	.950
0.6	1.175
0.7	1.375
0.8	1.600
0.9	1.775
1.0	1.975
1.1	2.175
1.2	2.375
1.3	2.600
1.4	2.825
1.5	3.050
1.6	3.300
1.7	3.550
1.8	3.800
1.9	4.075
2.0	4.325
2.1	4.750
2.2	4.900
2.3	5.200
2.4	5.500

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TABLE I. Time of flight for range settings (cont'd).

Range (KM)	Time (Sec)
2.5	5.775
2.6	6.100
2.7	6.425
2.8	6.725
2.9	7.075
3.0	7.375
3.1	7.725
3.2	8.075
3.3	8.425
3.4	8.800
3.5	9.150
3.6	9.525
3.7	9.900
3.8	10.275
3.9	10.650
4.0	11.050
4.1	11.425
4.2	11.850
4.3	12.275
4.4	12.700
4.5	13.125
4.6	13.550
4.7	13.975
4.8	14.400
4.9	14.825
5.0	15.275
5.1	15.725
5.2	16.225
5.3	16.700
5.4	17.200
5.5	17.700
5.6	18.200
5.7	18.700
5.8	19.225
5.9	19.725

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TABLE I. Time of flight for range settings (cont'd).

Range (KM)	Time (See)
6.0	20.275
A	7.375 (3.0KM Default Condition)

3.4.3.1.1.3 Penetration selector. A single thumbwheel switch shall be provided for selecting penetration after impact for use with XM433 RC fuzes. Time of function shall be taken from the FCC. In the absence of an FCC, the time of function shall be as shown in Table II.

TABLE II. Penetration/time conversion.

Penetration (M)	Time (M Sec)
SQ	0
BNK	20
10	30
15	110
20	200
25	325
30	450
35	525
40	575
45	600

3.4.3.1.1.4 Rate selector. The rates of fire, selectable by means of the selector switch, shall be: slow (180 ± 18 milliseconds), fast (60 ± 6 milliseconds for non-RC fuzed rounds and XM433 RC fuze, 70 ± 7 milliseconds for XM439 RC fuzed rounds), and automatic rate, depending on type of warhead, as listed in paragraph 3.4.3.1.1.1.

3.4.3.1.1.5 Mode selector. By means of the Front Panel mounted mode selector switch, the DU shall be capable of providing the following modes of fire:

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SNG Single fire - One round from one launcher, successive firings shall alternate between launchers from side to side to maintain weight balance.

PRS Pair Fire - Two rounds, one from each side of the aircraft.

QAD Quad Fire - Four rounds, one from each launcher with the inboard rockets leading the outboard rockets by 30 ± 5 milliseconds.

The DU shall revert to the next lower mode of fire if rockets are not available for the mode selected.

3.4.3.1.1.6 Quantity selector. By means of the Front Panel mounted Quantity selector switch, the DU shall be capable of selecting the Quantity of Rockets to be fired. The DU shall cause the preselected number of rockets to be fired with each depression of the wing stores firing switch as long as the switch is held. If the wing stores firing switch is released before the required quantity is released, the DU shall cease firing within one firing cycle. A successive depression of the wing stores firing switch shall begin a new firing sequence.

3.4.3.1.1.7 Zone arming switches. The five zone arm pushbutton switches provide the means of arming the zone for a particular type of warhead. Zones shall be armable if the master arm switch is in the "ARM" position and a zone arming switch is depressed. A successive depression of a zone arm switch will cause the previously armed zone(s) to be disarmed and zone selected if previously unarmed to become armed. This action may be performed while firing or not firing rockets. The depression of a zone arm switch shall arm or disarm all zones having the same warhead type selection.

3.4.3.1.1.8 Test switch. The DU provides a single press-to-test switch for the purpose of executing a complete built-in test function of the RMS. The test shall only be executable with the master arm switch in the standby position. The test may be initiated while on the ground or in flight. During the test the following visible action shall take place.

1. All display indicators shall be lit for approximately 3 seconds.

2. The display shall indicate the status of each LRU within the RMS for approximately 3 seconds in order from left to right, left outboard OU, left inboard OU, DU, right inboard OU, right outboard OU. A good LRU shall be signified by the presence of a "7" in the zone display, a failed unit shall be indicated by a

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blank display. A blank in the DU position shall indicate a failure in the FCC communications link, as no other "soft" failure shall exist. A "hard" failure of the DU shall cause an all "8" indication on the display and all controls to be non-functional. Power may be recycled to the DU and the test repeated.

3.4.3.1.1.9 Display. The DU provides a rounds remaining display on which the number of rockets remaining in each zone is displayed. The lighting intensity of the display shall be controlled by the helicopter panel lighting circuitry and shall be compatible with both daylight and night vision modes of operation. Characters shall be discernable in a 100,000 foot-candle environment with the control set to full brightness or if the control voltage is less than 0.6 VDC (panel lighting control "off"). Display color shall be aviation red per MIL-C-6781. The ten's digit of each zone display shall automatically turnoff when the count is less than 10. In the night vision mode the display shall be compatible with the infrared night vision goggles AN/PVS-5.

3.4.3.1.1.10 Panel lighting. Descriptors and switch legends shall be lighted aviation red when power is applied and the lighting control voltage is applied. When not lit, such legends shall appear as white lettering on a black background. Edge lit panels shall conform to MIL-P-7788, Type IV or Type V. Dimming shall be controlled by the helicopter light dimming control. The edge lit panel shall be compatible with AN/PVS-5 night vision goggles.

3.4.3.1.1.11 Zone arm indicators. The DU shall provide zone arm indicators to display the arming status of each zone. This indicator shall not be visible when the zone is disarmed except during self-test and shall be a lighted bracket on the zone arm switch when the zone is armed. The indicators shall be an integral part of the zone-arm pushbuttons and shall be relampable without disassembly of the unit.

3.4.3.1.2 Electrical interface. All electrical interface with the DU shall be through one 100-contact connector mounted on the back of the DU. The DU connector shall provide all interface between the helicopter and the RMS. Helicopter interface shall include standby power, master arm power, lighting control, firing signal, and fire control system. The DU shall provide power and serve as the master control unit for up to four OU's within the RMS. Connector pin assignments shall be as shown in Table III.

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TABLE III. Display unit connector J1 (cont'd).

Pin #	Signal Name	Function	Logic "0" V Max	Logic "0" Current	Logic "1" Volts(Min)	Logic "1" I Max
1	A0	ADDRESS LINES (OUTPUT)	0.4V	10 ma	2.4V	-11ma
2	A1					
3	A2					
4	A3					
5	A4					
6	A5					
7	A6					
8	A7					
9	A8					
10	A9					
11	A10					
12	A11					
13	A12					
14	A13					
15	A14					
16	A15					
17	D0	DATA IN/OUT	.95(IN) .45V(OUT	-0.25 ma 25 ma	2.0V(IN) 2.4V(OUT)	200μa -10 ma
18	D1					
19	D2					
20	D3					
21	D4					
22	D5					
23	D6					
24	D7					
25	NSHTDN	SHUTDOWN(OUT)	0.4V	2.6 ma	3.5	-380μa
26	NCS1	CIRCUIT SEL ACTIVE LO OUT	0.4V	3.2 ma	2.5	-360μa
27	NCS0					
28	NCS3					
29	NCS2					
30	NKDOG	WATCHDOG TRIG (OUT)	0.4V	3.6 ma	2.5	-380μa
31	NCNTRO	CTROL STROB (OUT)				
32	NSTRD	STATUS STROBE (OUT)				
33	NHLD	HOLD ACTIVE LO (IN)	0.8V	-1 ma	3.3V min	0
34	BUSEN	BUS ENABLE	0.8V	-1.4 ma	2.4V min	0
35	NC					
36	OU CLK	2.048 MHz CLK OUT	0.45	1.0 ma	3.7V	130μa

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TABLE III. Display unit connector J1 (cont'd).

37	NMRD	NOT MEM READ (OUT)				
38	NMWR	NOT MEM WRITE (OUT)	-0.45	8 ma	2.4	-900μa
39	NIORD	NOT I/O READ (OUT)				
40	NIOWR	NOT I/O WRITE (OUT)				
41	NINIT	NOT INITIALIZE (OUT)	0.6V	1 ma	5.0V	Z_o=1K
42	TMR	200 Hz CLK (OUT)	0.4	3.6 ma	2.4V	-380μa
43	ACK	ACKNOWLEDGE (IN)	0.8V	-1 ma	3.3V	0
44	HLDA	HOLD ACK (OUT)	0.45V	1.0 ma	3.3V	150μa
45	NC					
46	NCPU	CPU I.D. (OUT	OV		OPEN	
47	NC					
48	TWC	THUMBWHEEL LAMP CONTR		0 to 10V		
49	VCON	10V PWR SUP CONTR		0 to 5V		
50	DIMRO	READOUT VOLTAGE	(ANALOG)	0 to 5V		
51	ROC	READOUT CONTR		0 to 1V		
52	DIMV	LAMP SUPPLY		0 to 10V		
53	NINTR	OUT/IN	0.4V	2 ma	2.4V	0 ma
54	NRST	NOT RESET	0.4V	50 ma	OPEN	
55	INTE	INTERRUPT ENABLE (IN)	0.8V		2.3V	
56	ROB					
57	RIB	OU ESI TIE				
58	LIB	POINTS (NOT				
59	LOB	USED)				
60	ESI		OPEN		+28VDC	0.5A
61	STBYO	+28V STANDBY			+28VDC	.100A
62	NC					
63	STBY	+28V STANDBY			+28V	3A
64	STBY	+28 STANDBY				
65	NTREN	MAP RLY ENABLE	1.0V	100 ma	+28V	
66	NXARM	ZONE ARM EXT (IN)	0.4V	0.6 ma	2.4V	0
67	NC					

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TABLE III. Display unit connector J1 (cont'd).

68	PRTN	PWR RETURN			OV	3A
69	PRTN					
70	PRTN					
71	GND					
72	NFSD	FCC SERIAL DATA	SEE PARA 3.4.3.1.11			
73	FSD					
74	NRSD	ROCKET SERIAL DATA	SEE 9324400			
75	RSD					
76	DOG	WATCHDOG T.P.	0.4V	3.60 ma	2.4V	380μa
77	VOUMON	OU MONITOR T.P.	0.4V	5 ma	5.0V	Z_o=6.8K
78	NSESI	SET ESI T.P.	1V	100 ma	+28V	
79	NRESI	RESET ESI T.P.	1V	100 ma	+28V	
80	XRESI	EXT ESI RESET (IN)	0.4V	0 ma	+5.0V	3 ma
81	STBYMON	+28V MONITOR	0.4V	5 ma	+3.0V	.5 ma
82	FCCCLK	153.6 KHz CLK	0.4V	3.6 ma	2.4V	380 μa
83	NC					
84	VOU	OU PWR			+20V MIN +21.5V MAX	2.85 A MAX
85	+5V	+5V TEST PT			5V+5%	
86	-5V	-5V T.P.			-5V+5%	
87	+12V	+12V T.P.			+12V+5%	
88	DIMC	DIMMER CONTR		0 to 28V @ 100 ma MAX		
89	FCCG	FCC GOOD (IN)	OPEN		+28V	7 ma
90	SHLD	CHASSIS				
91	CHASSIS	GND				
92	FCCRTN	GND				
93	MAPO	MASTER ARM			28V	16A PEAK
94	MAPO	POWER TO OU				
95	NC					
96	MAP	MASTER ARM				
97	MAP	PWR	+28VDC	16A MAX PEAK	10 M SEC	
98	NC					
99	NC					
100	TRIG	TRIGGER INPUT	+28VDC @ 100 ma			

3.4.3.1.3 Operational control. The DU shall control the operation of the four OU's through a twisted pair serial data line as defined by the DU/OU Interface Specification, 9324400. The operations include taking inventory, conducting self-test, charging fuzes, and firing rockets.

3.4.3.1.4 Accuracy. The accuracy for both the manual and automatic setting modes, excluding fire control errors shall be ±1 millisecond in the data sent to the OU's.

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3.4.3.1.5 Distribution of weight. The distribution of weight between right and left launchers shall not be out of balance by more than five rockets except for asymmetrical loading. The firing order of each type of rocket within a launcher shall be controlled to minimize lateral center of gravity shifts within the launcher.

3.4.3.1.6 Launch delay. The DU shall complete the fire command sequences to the first OU pair within 50 milliseconds after the wing stores firing switch is depressed.

3.4.3.1.7 Failsafe. Failure of any assembly shall not cause firing of any store.

3.4.3.1.8 Fault detection/location. The DU shall have a Built-In-Test (BIT) capability to check the operational status of the equipment. The fault detection/location function shall be accomplished when the STANDBY/ARM switch is in the STANDBY position. Arming of the RMS shall override self test. BIT shall obviate the need for any external test equipment for the performance of preflight, or in-flight testing, alignment of the RMS and fault isolation down to the defective line replaceable unit (LRU). All RMS functions should be tested and the BIT provisions shall functionally test the RMS from end-to-end, except for the firing circuits.

3.4.3.1.9 Malfunction indication. The equipment status indicator a GO/NO GO device, shall be capable of holding the last result if the power is interrupted or removed. The output of the BIT, which drives a remote indicator, shall be such that a signal of 28 ± 4 VDC indicates a go condition and a signal of ± 0.5 VDC indicates a no go condition. Failure of the DU other than an FCC communications failure shall result in a shutdown of the DU. This failure shall be indicated by all "8's" on the display.

3.4.3.1.10 Failure of OU. Upon failure of an OU the DU shall disable from firing the corresponding OU on the opposite side of the aircraft in order to preserve weight balance.

3.4.3.1.11 Fire control. The DU shall provide rocket fuze/warhead type selection and fuze setting functions to the fire control computer. The DU shall have the capability to set RC airburst and penetration type fuzes with or without the fire control computer (FCC). FCC range data shall be used only if the automatic range mode is selected. The DU shall accept time of function data from the FCC. If the FCC is not present, or if setting airburst fuze in other than the automatic setting, the time of function shall be selected from a stored conversion table based on a 60 knot curve. (Reference Tables I and II.) Data input and output shall be received and transmitted in accordance with the Signal Interface Control Document 12011837.

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3.4.3.1.12 Power requirements. The DU shall operate in accordance with MIL-STD-704A, Category B, Abnormal Steady state Level (ASSL) 1. Normal operating voltage shall be ± 28 volts DC and average current shall not exceed 3 amperes. Reversal of polarity of the primary power shall not damage the DU or other RMS components, or cause false fuzing or firing signals. There shall be no false outputs resulting from transients or from application or removal of power.

3.4.3.1.12.1 Standby power. Standby power, which provides the primary power for the RMS, shall be supplied through the helicopter master arm switch.

3.4.3.1.12.2 Master arm power. Arming power shall be controlled by the master arm switch, for the purpose of powering rocket fire circuits. The DU shall control the power to the operations units such that no arming power is provided unless a zone arming function is executed.

3.4.3.1.12.3 DU Panel illumination control/power. The DU shall interface with the helicopter lighting circuitry for electrical control and power to illuminate the panel lights and display.

3.4.3.1.13 Internal power supplies. For over-voltage protection, the voltage trip point shall be less than the maximum voltage rating of the devices supplied by the internal power supplies.

3.4.3.1.13.1 Operations unit power. This secondary power source, derived from standby power, shall be regulated with over-voltage and short circuit protection. The voltage available at the DU output terminals shall be 20 VDC minimum, 21.5 VDC maximum with an output current capability of 2.0 amperes steady-state, 2.85 amperes peak for 3 seconds maximum.

3.4.3.1.14 Trigger input. The DU shall interface with the wing stores firing switches on the cyclic controls. Closure of the wing stores firing switches shall be the final step in the rocket firing sequence.

3.4.3.1.15 Mechanical interface. The panel shall be in accordance with MS25212. The DU shall not exceed 4.5 inches in height and 7.0 inches in length. The outline of the DU shall conform to 9324104, DU Interface Control Drawing.

3.4.3.1.16 Proof test. Display Unit assemblies shall operate when assembled to an accepted AH-1S helicopter.

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3.5 Environmental conditions. The Display Unit shall operate and meet the performance requirements of 3.4.3 under any of the environmental conditions, or reasonable combination of these conditions, as specified below:

3.5.1 Natural environments. The Display Unit shall be capable of operation and storage in the worldwide environments as specified below:

3.5.1.1 Altitude. Operating: 0-20,000 feet; nonoperating and storage: 0-50,000 feet.

3.5.1.2 Temperature.

Display Unit.

a. Operating: -65° to +125°F
b. Nonoperating and storage: -65° to +160°F

3.5.1.3 Humidity. Operation after exposure to 149°F and relative humidity to 95 percent.

3.5.1.4 Sand and dust. Operation after internal and external exposure to sand and dust test.

3.5.1.5 Fungus. Operating, nonoperating and storage after being subjected to fungus specified below:

Fungus	Serial Number
Aspergillus niger	9642
Aspergillus flavus	9643
Aspergullus versicolor	11730
Penicillium fumculosum	11797
Chaetomum globosum	6205

3.5.1.6 Explosive atmosphere. Operation in explosive vapor within or surrounding the Operations Unit.

3.5.1.7 Service environment. The Display Unit shall be designed and constructed to withstand the service environment as specified in MIL-E-5400 for Class 1A equipment.

3.5.1.7.1 Vibration. The Display Unit shall not sustain damage or subsequently fail to provide specified performance when subjected to the vibration environment defined by MIL-E-5400, Figure 2, (Sheet 1), Curve III.

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3.5.1.7.2 Shock. The Display Unit shall meet the shock requirements of paragraph 3.2.24.6.1 of MIL-E-5400. The requirements of paragraph 3.2.24.6.2 of MIL-E-5400 for mounting hardware crash safety shall only apply to the units mounted inside the cockpit.

3.5.1.7.2.1 Bench handling. The Display Unit shall withstand shock environments encountered in maintenance and service.

3.6 Workmanship.

3.6.1 Fabrication. Finished items and parts shall be fabricated in a thorough, workmanlike manner, and all manufacturing, processing and assembly operations shall be correctly performed. Unauthorized salvaging operations (e.g., hammering to shape, repair by welding, straightening, bending, etc.) shall not be permitted.

3.6.2 Parts. The parts shall not exhibit faulty material and processing such as seams, laps, laminations, cracks, visible steps, sharp edges, nicks, scratches, burrs, deformations, unblended radii, chips, dents, rust, dirt, grease and oil (except where specifically required), corrosion products, flux, other foreign matter, and missing operations which may affect serviceability, operation, or safety.

3.6.3 Fasteners. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.4 Cable assemblies and wires. Cable assemblies and wires shall be neatly dressed and tied in such a manner as to prevent pinching or chafing the wire insulation. Ends of wires shall be free from burrs, sharp edges or points.

3.6.5 Castings. Casting shall be clean, sound, and free from blowholes, porosity and inclusion.

3.6.6 Welding. Welds shall be reasonably smooth and free of craters and cracks. They shall exhibit characteristics of good fusion penetration and soundness of weld deposit representative of good welding practice. They shall be free of irregularities which indicate lack of skill or experience on the part of the operator. All scale and splatter shall be completely removed prior to application of protective coatings. There shall be no cracks in or near welded areas.

3.6.7 Soldering. All soldering shall be performed in a thorough, workmanlike manner with no evidence of burning or charring.

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3.6.8 Coatings. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted.

3.6.9 Cleaning. The cleaning agents shall not be injurious to any part or assembly.

3.6.10 Packing. All required packing shall be dry.

3.6.11 Markings. All required markings and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-W-63150 shall apply and are hereby made a part of this detail specification. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Certification of provisions. The certification provisions of MIL-W-63150 shall apply.

4.3 Classification of inspection. The following types of inspection shall be conducted on this item.

a. First Article Inspection.

b. Quality Conformance Inspection.

4.4 First article inspection.

4.4.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.4.2. The first article sample shall consist of five Operation Units.

4.4.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government, at a place designated by the Government, to any or all of the examinations and tests specified in the item detail specification and to any or all of the requirements of the applicable drawings. See Table IV specified herein.

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4.4.3 Rejection. If any assembly, component, or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or test specimen in the sample to comply with any of the stated requirements.

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TABLE IV. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Display Unit				See Below
					NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE *UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Display Unit (Dwg 9324107)				
	Examination for defects	5	3.4	4.5.2.1	
	Interchangeability	5	3.4.1	4.6.3	
	Proof test	5	3.4.3. 1.16	4.6.4	
	Altitude	5		3.5.1.1	4.6.5.1
	Temperature	5		3.5.1.2	4.6.5.1
	Humidity	5		3.5.1.3	4.6.5.2
	Sand and dust	5		3.5.1.4	4.6.5.3
	Fungus	5		3.5.1.5	4.6.5.4
	Explosive atmosphere	5		3.5.1.6	4.6.5.5
	Service environment	5		3.5.1.7	4.6.5.6
	Vibration	5		3.5.1. 7.1	4.6.5.6.1
	Shock	5		3.5.1 7.2	4.6.5.6.2
	Bench handling	5		3.5.1. 7.2.1	4.6.5.6.2.1

NOTES:
* The same five units shall undergo all the Examination or Test in the order listed.

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4.5 Quality conformance (acceptance) inspection. Acceptance inspection shall be performed and shall be in accordance with the following:

4.5.1 Lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity critieria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with the requirements.

4.5.2 Examination. Examination for the inspection listed in the following Classification of Defects and Tests paragraphs shall be performed by sampling to the AQL specified unless 100 percent inspection is indicated. Unless otherwise specified sampling shall be in accordance with MIL-STD-105 Level II, except the contractor may request permission from the procuring activity to use an equivalent continuous sampling plan from MIL-STD-1235.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70484(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.5.2.1	Display Unit		9324107
			NEXT HIGHER ASSEMBLY 9324106

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Operations unit damaged (puncture, dented or deformed)		100%	3.4	Visual
102.	Assembly incorrect		100%	3.4	Visual
103.	Assembly incomplete		100%	3.4	Visual
104.	Component missing, damaged or improperly assembled		100%	3.4	Visual
105.	Operation missing or improperly performed		100%	3.4	Visual
106.	Nameplate label missing		100%	3.4	Visual
107.	Materials not as specified		100%	3.4	Visual
108.	Preconditioning/Performance		100%	3.4.2/ 3.4.3	4.6.2
Minor					
201.	Nameplate marking misleading, illegible, incomplete or missing		1.5%	3.4	Visual
202.	Poor workmanship		2.5%	3.6	4.6.6

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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4.5.3 Testing. The tests (First Article and Quality Conformance) listed herein, shall be performed on inspection lots as defined in 4.5.1 of this specification. The sampling plans shall conform to the provisions of 4.5.2 above.

4.5.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection column in the tables, starting with paragraphs 4.5.2.1 and test methods paragraph (see 4.6). The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See 6.3 herein.

4.6 Methods of inspection.

4.6.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, sub-assemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.6.1.1 Visual and mechanical inspection. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.5.2 and Table IV.

4.6.2 Performance test. The Display Unit shall be tested for compliance with specified performance requirements after being subjected to the preconditioning requirement (see 3.4.2). Any M138 Display Unit which fails to comply with the specified requirement or evidence of degradation of its operation shall be classed defective. Test equipment and procedure shall be satisfactory to the contracting officer (see 4.5.4).

4.6.3 Interchangeability (first article inspection only). The Display Unit shall be subjected to the interchangeability tests specified (see 3.4.1). at the discretion of the Government representative, an interchangeability retest may be allowed. Failure during the retest shall cause rejection of the first article. A sample of five Operation Units from each retest shall be tested using the same procedures specified.

4.6.4 Proof test (first article inspection only). When specified in the contract (see 6.2) one (1) M138 Display Unit shall be forwarded to the site specified by the contracting agency and utilized on an AH-1S helicopter to determine that the test set will function properly within the system. The test sets shall function in accordance to the intent of the design and with no evidence of malfunction.

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4.6.5 Environmental tests (first article inspection only).

4.6.5.1 Temperature - altitude. The Operations Unit shall be subjected to the test of Method 504.1 of MIL-STD-810 for category 3 equipment with the following exceptions;

a. Use 20,000 feet where 15,000 feet is specified.

b. Do not exceed the temperature limits of 3.5.1.2.

4.6.5.1.1 High temperature. The high temperature test requirements are satisfied by 4.6.5.1. No separate test is required.

4.6.5.1.2 Low temperature. The Operations Unit shall be subjected to the test of Method 502.1, Procedure I of MIL-STD-8l0, except maximum low temperature to be -65°F.

4.6.5.2 Humidity. The Operations Unit shall be subjected to the test of Method 509.1 of MIL-STD-810.

4.6.5.3 Sand and dust. The Operations Unit shall be subjected to the test Method 510.1 of MIL-STD-810, Procedure I, with the exception that for equipment mounted enternal to the aircraft, the air velocity in Steps 1, 2 and 3 shall be 200 ± 100 fpm. Step 3 may proceed immediately after stabilization in Step 2, if desired.

4.6.5.4 Fungus. The Operations Unit shall be subjected to the test of Method 508.1 of MIL-STD-810, Procedure I. Documentary evidance acceptable to the procuring activity that the material will not be nutrients for fungi may be used in place of the tests.

4.6.5.5 Explosive atmosphere. The Operations Unit shall be subjected to the test of Method 511.1 of MIL-STD-810, Procedure I, except test to be limited to maximum altitude of 20,000 feet.

4.6.5.6 Service environment. The Operations Unit shall be tested in compliance with the requirement of section 3 of this specification in accordance with 3.5.1.7.

4.6.5.6.1 Vibration. The Operations Unit shall be subjected to Method 514.2-3 of MIL-STD-810, Procedure I, Category C.

4.6.5.6.2 Shock. The Operations Unit shall be subjected to the Test Methods of MIL-STD-810, Procedure I.

4.6.5.6.2.1 Bench handling. The Operations Unit shall be subjected to the Test Method 516.2 of MIL-STD-810, Procedure I.

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5. PACKAGING

5.1 Preservation, packaging, packing and marking. The display unit shall be preserved, packaged and marked in accordance with the requirements of the Preservation and Packaging Data Sheet for P9324107 at the level of protection specified in the contract (see 6.2b).

5.1.1 Spare parts or replacement parts. Spare parts or replacement parts shall be cleaned, preserved, and packaged as specified in the contract or order.

5.2 Commercial packaging shall be in accordance with ASTM-D-3951.

6. NOTES

6.1 Intended use. The Display Units are intended to be used in combination with two or four Operations Units to form the M138 Rocket Management Subsystem for controlling the fuze setting and firing of 2.75 Inch Aerial Rockets.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Serialization requirements, if applicable.

d. Quality Conformance Inspection, if other than specified in Section 4 of this specification.

e. First Article sample requirements, if other than specified in Section 4 of this specification.

f. Packaging requirements, if other than specified in Section 5 of this specification.

g. Certificates of conformance for each lot or shipment of product.

h. Location to which items are to be shipped.

6.2.2 Contract data requirements. Contract data requirements for inspection equipment design shall conform to Data Item Description DI-R-1714, Tailored.

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6.2.2 Contract data requirements. Contract data requirements for inspection equipment design shall conform to Data Item Description DI-R-1714, Tailored.

6.3 Submission for inspection equipment design approval. See MIL-W-63150. Equipment designs shall be submitted as required to: Commander, AMCCOM, ATTN: AMSMC-QAF-I(D), Dover, NJ 07801-5001. Letter of submittal shall state contractor, contract number, specification number, item nomenclature, and classification of defects and test paragraphs .

6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to: Commander, AMCCOM, ATTN: AMSMC-QAF-S(D), Dover, NJ 07801-5001.

6.5 Drawings. Drawings listed in Section 2 of this specification, under the heading US Army Armament Research and Development Center (ARDC), may also include drawings prepared by and identified as US Army Armament Research and Development Command (ARRADCOM), Rock Island Arsenal or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDC.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-70484	DISPLAY UNIT, FOR ROCKET MANAGEMENT SUBSYSTEM M138

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Includ Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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